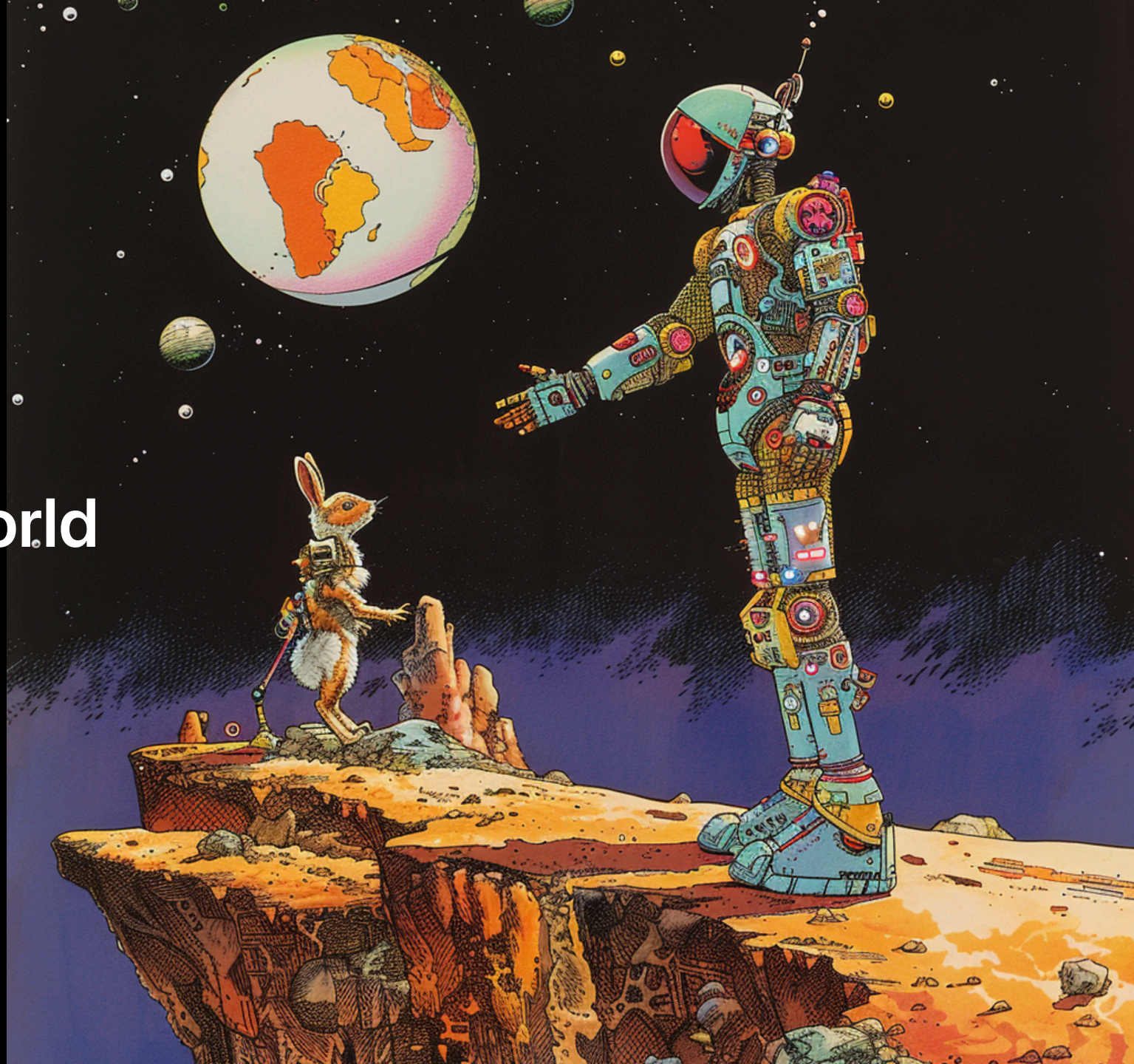


Hello World



Intro Kickstart-kursus i programmering 2024

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DIKU Københavns Universitet



The Team

- Daniel Spikol
- Alexander Mittet
- Thea Nyks Meggele
- Jeppe Fræhr Møller
- Mathilde Kjer Parlo
- Andrea Sebro Rasmussen
- Casper Brahe Rützou
- Viktoria Silke Sofia Strandbygaard
- Lucas Østergaard Jarmer

Inclusive DIKU

- Welcome to DIKU. Our goal is to promote excellence in computer science education and research.
- To strive for a respectful, inclusive, diverse environment and encourage open and critical academic discussion.
- We strive to create a welcoming and respectful environment.

OPEN & ACCOMMODATING

Being open and respectful of differing opinions, viewpoints, and experiences



INCLUSIVE

Focusing on what is best not just for us as individuals, but for the overall community

CONSTRUCTIVE

Giving and accepting constructive feedback



RESPONSIBLE

Accepting responsibility and apologizing to those affected by our mistakes, and learning from the experience

RESPECTFUL

Respecting DIKU's facilities by following the current regulations of the Faculty of Science and DIKU



Velkommen!

- Presentation of Activities
- Why and what is the Kickstart course?

Overall Plan for the week

Day 1 Mon

6h 00m

🔒 09:00

Welcome and Introducti...

15m

09:15

Greeting Activity

15m

09:30

Team Building and Icebr...

1h 30m

11:00

Installation, wifi, setup, ...

45m

11:45

Lunch

45m

12:30

Intro to Processing

10m

12:40

Worksheet: #0 The First...

1h 00m

13:40

Worksheet: #1

1h 20m

15:00

+ Add block

Day 2 Tues

6h 00m

🔒 09:00

Hello and Intro to peer p...

15m

09:15

Intro to setup/draw + gl...

15m

09:30

Break

15m

09:45

Worksheet #2

1h 00m

10:45

Worksheet #3

1h 00m

11:45

Lunch

45m

12:30

Pair Programming Practi...

15m

12:45

Worksheet: #3

1h 15m

14:00

Break

15m

14:15

Peer Feedback

30m

🔒 14:45

Coding and end of day

15m

15:00

+ Add block

Day 3 Weds

6h 00m

🔒 09:00

Intro talk on Conditionals

30m

09:30

Break

10m

09:40

Worksheet #4 and #5

1h 50m

11:30

Peer Feedback Practice

15m

11:45

Lunch

50m

12:35

Projects and Brainstorm...

30m

13:05

Project Time Brainstorm...

1h 55m

15:00

+ Add block

Day 4 Thurs

6h 00m

🔒 09:00

Creative Coding Valkari...

30m

09:30

Worksheets #5-6 & Proj...

1h 00m

10:30

Break

15m

10:45

Worksheets

1h 15m

12:00

Lunch

1h 00m

13:00

Project Time and/or Wor...

20m

13:20

Project Time and/or Wor...

40m

🔒 14:00

break

10m

14:10

Project Time and/or Wor...

50m

15:00

+ Add block

Day 5 Fri

6h 00m

🔒 09:00

Summary of week

45m

🔒 09:45

Project time

2h 15m

🔒 12:00

Lunch

45m

12:45

Final Project Prep

45m

13:30

Presentations

1h 15m

14:45

Survey

15m

15:00

+ Add block

Monday's Plan

- kl. 9:00-9:45 → Welcome and Introduction
- kl. 10:00-11:15 → Team Building Activities, Mini-Lecture, and Coding
- KL 11:15-12:00 → Processing Introduction
- kl. 12:00 - Lunch → Options (Bio Kantine...)
- kl. 13:00-13:30 → Aud.06 Intro to Processing
- kl. 13:30-15:00 → Team rooms - Intro to Processing
- kl. 14:45-15:00 → Check-in with your Teams

Intended Fun Outcomes

- Why IFOs and not Intended Learning Outcomes
 - Intended Learning Outcomes (ILOs) define what a learner will have acquired and will be able to do upon completing their courses and studies.

Intended Fun Outcomes

- Meet people and collaborate
- Explore programming with a focus on creating games
- Have fun and take chances through creativity
- Take control of your education

Meet people and collaborate

- Design Thinking
- Collaboration
- Reframing



Teams and Instructors

- **Team Alpha:** Alexander, Viktoria, and Thea → UP1 workroom 1
- **Team Bravo:** Mathilde and Jeppe → UP1 workroom 2
- **Team Charlie:** Casper, Lucas, and Andrea → UP1 workroom 3

Challenge Time

- Design Thinking
- Collaboration
- Reframing

Time for a Team Challenge



Marshmallow Challenge Reflection

- [TED TALK Tom Wujec](#)
- Constant prototyping as a problem-solving method.
- **Getting the Design Right and the Right Design** → Buxton, William (2007). *Sketching user experiences: getting the design right and the right design*. Amsterdam: Elsevier/Morgan Kaufmann

Internet Access

Aktiver KU-bruger

- Log på mit.ku.dk med NemID og find midlertidig pinkode. Gå til kunet.dk og tryk "Første gang du logger på".
- Log på med den midlertidige kode og CPR nummer uden bindestreg.
- Aflæs KU brugernavn (fx abc123) og angiv password.

Eduroam WiFi

- Kræver KU bruger og password
- Log på med brugernavn: abc123@ku.dk

Alternativt: KU Guest WiFi

- Opret 24 timers konto ved at indtaste navn, mail, mobilnummer.
- Modtag kode via SMS.

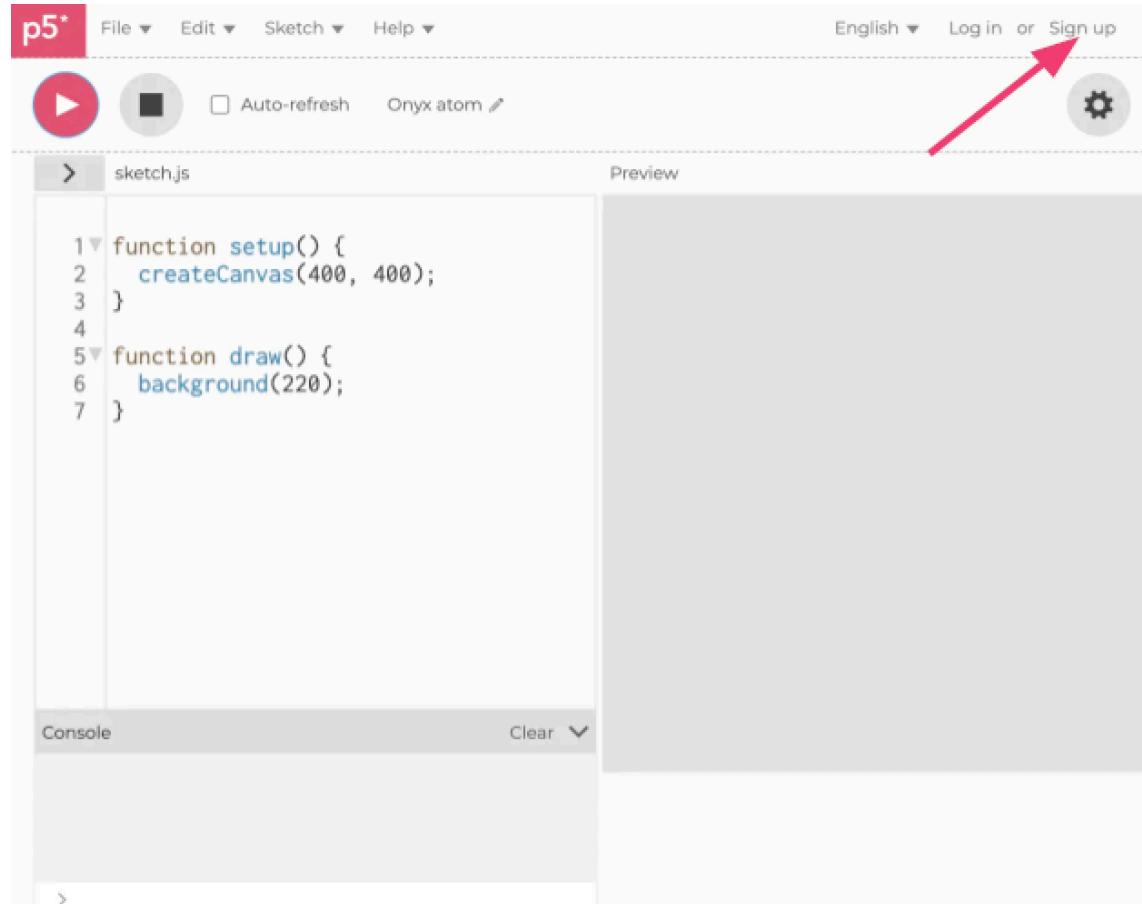
Installation af p5*js

<https://p5js.org/>



Web Editor Steps

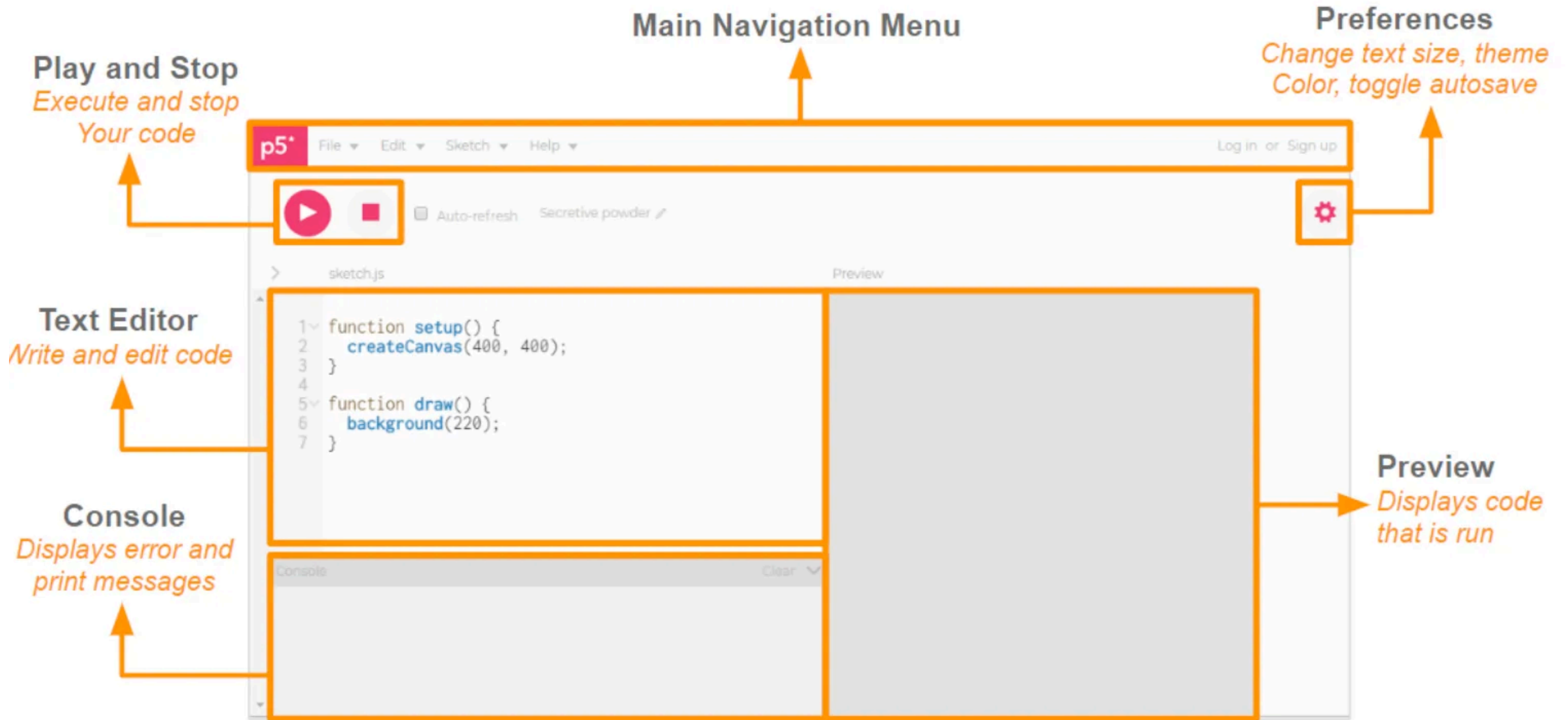
1. Access to the Internet  and web browser 
2. The p5.js Web Editor is a website where programmers can write, test, share, or remix p5.js programs without needing to download or configure a code editor on a computer. A code editor makes writing and reading code easier by organizing and coloring text in a way that helps programmers see different parts of the code.
3. See this [Tutorial](#) for getting up to speed if you want.

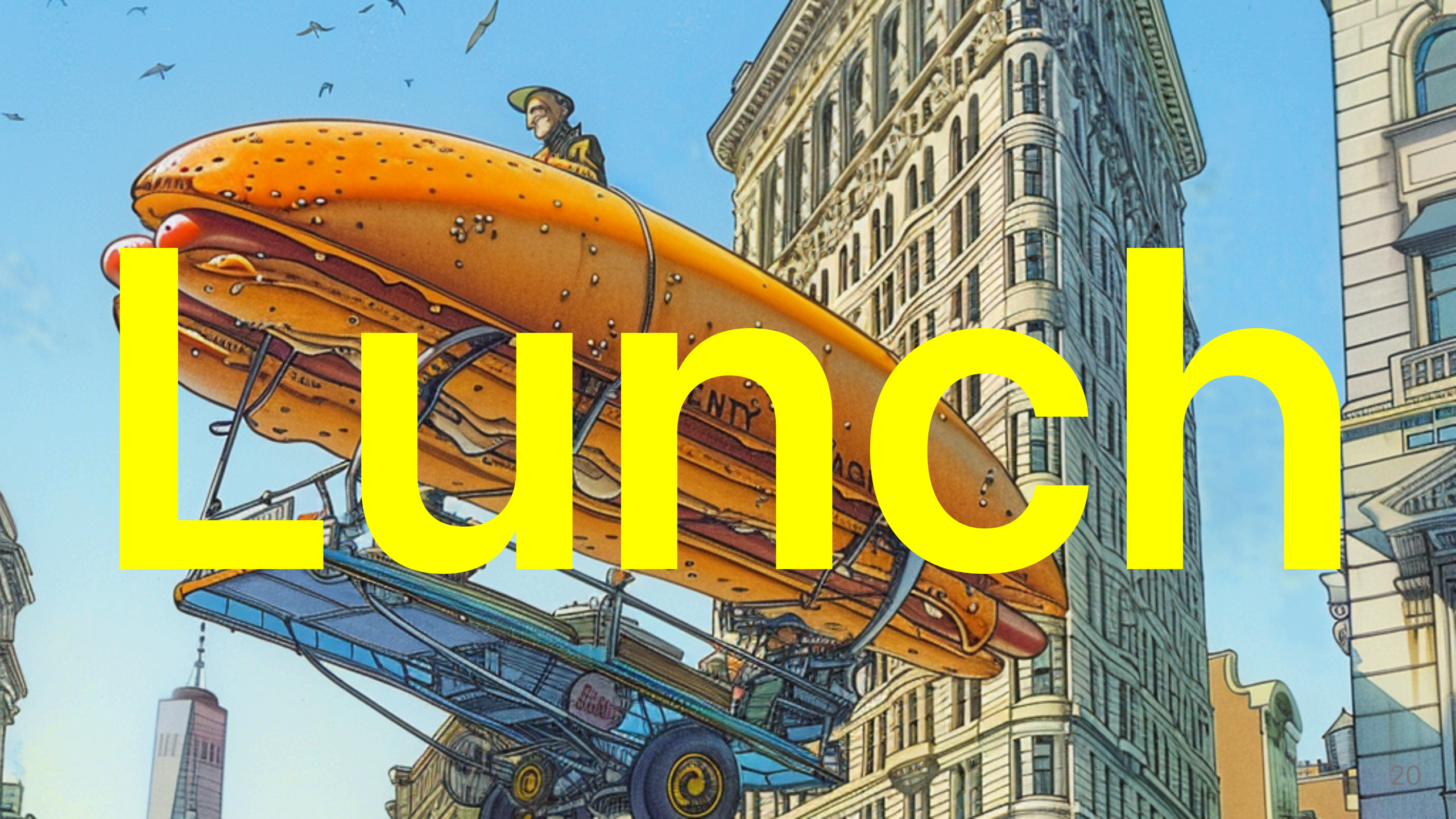


P5 Sign up

1. You will want to save and share your files
2. You can sign-up with your Email, Google, or GitHub

p5.js Web Editor Interface

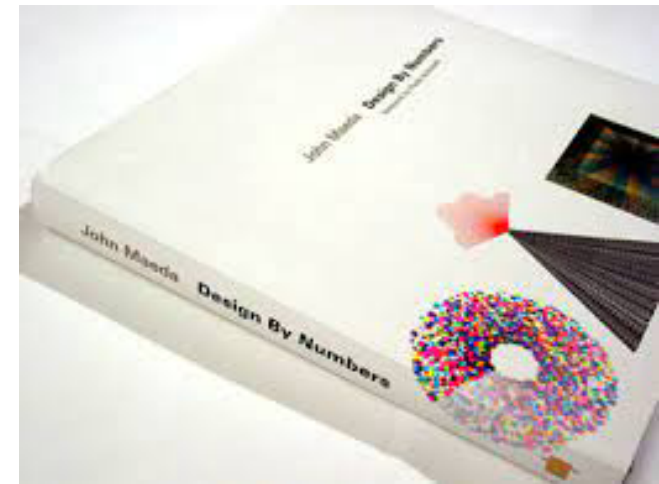




Lunch

Processing History

- M.I.T. Media Lab Casey Reas & Ben Fry
- Design By Numbers - Maeda, John. - Design by numbers / John Maeda.. - 1999. - ISBN: 0262133547
- Creative Programming



What is Processing

- Processing is a flexible software sketchbook and a language for learning code. Since 2001, Processing has promoted software literacy within the visual arts and visual literacy within technology. Thousands of students, artists, designers, researchers, and hobbyists use Processing for learning and prototyping.
- Processing Development Environment → PDE
- Programs are called Sketches
- The PDE consists of a simple text editor for writing code, a message area, a text console, tabs for managing files, a toolbar with buttons for common actions, and a series of menus.

Processing Example

p5.js Web Editor | cyclops02

https://editor.p5js.org/spikol/sketches/LctYxoL5V

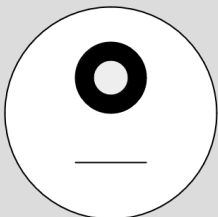
p5* File Edit Sketch Help English Hello, spikol!

Auto-refresh cyclops02 by spikol

sketch.js Saved: 1 day ago Preview

```
1 function setup() {
2
3   createCanvas(400, 400); // size of window
4   background(220); // canvas color
5
6   ellipse(200, 200, 150, 150); // the head
7
8   //draws the black eye
9   fill(0);
10  ellipse(200, 175, 50, 50);
11
12  //draws the pupil
13  fill(240);
14  ellipse(200, 175, 25, 25);
15
16  // draws the mouth
17  line(175, 235, 225, 235);
18 }
19
20 function draw() {
21   //fill(random (255), random (255), random (255));
22   //ellipse(200, 175, 25, 25);
23 }
```

Console Clear



Processing Parts 1

1. Basics of Processing IDE:

- **Interface:** The Processing IDE has a simple interface. There's an area for writing code, buttons for running and stopping sketches, and a message area below for error messages and other notifications.
- **Sketch:** In Processing, each project is called a "sketch". A sketch is a combination of code, data, and output.

2. **Language:** Processing uses a variant of the Java language. It's designed to be beginner-friendly, with simpler functions and setup to create visual and interactive projects quickly.

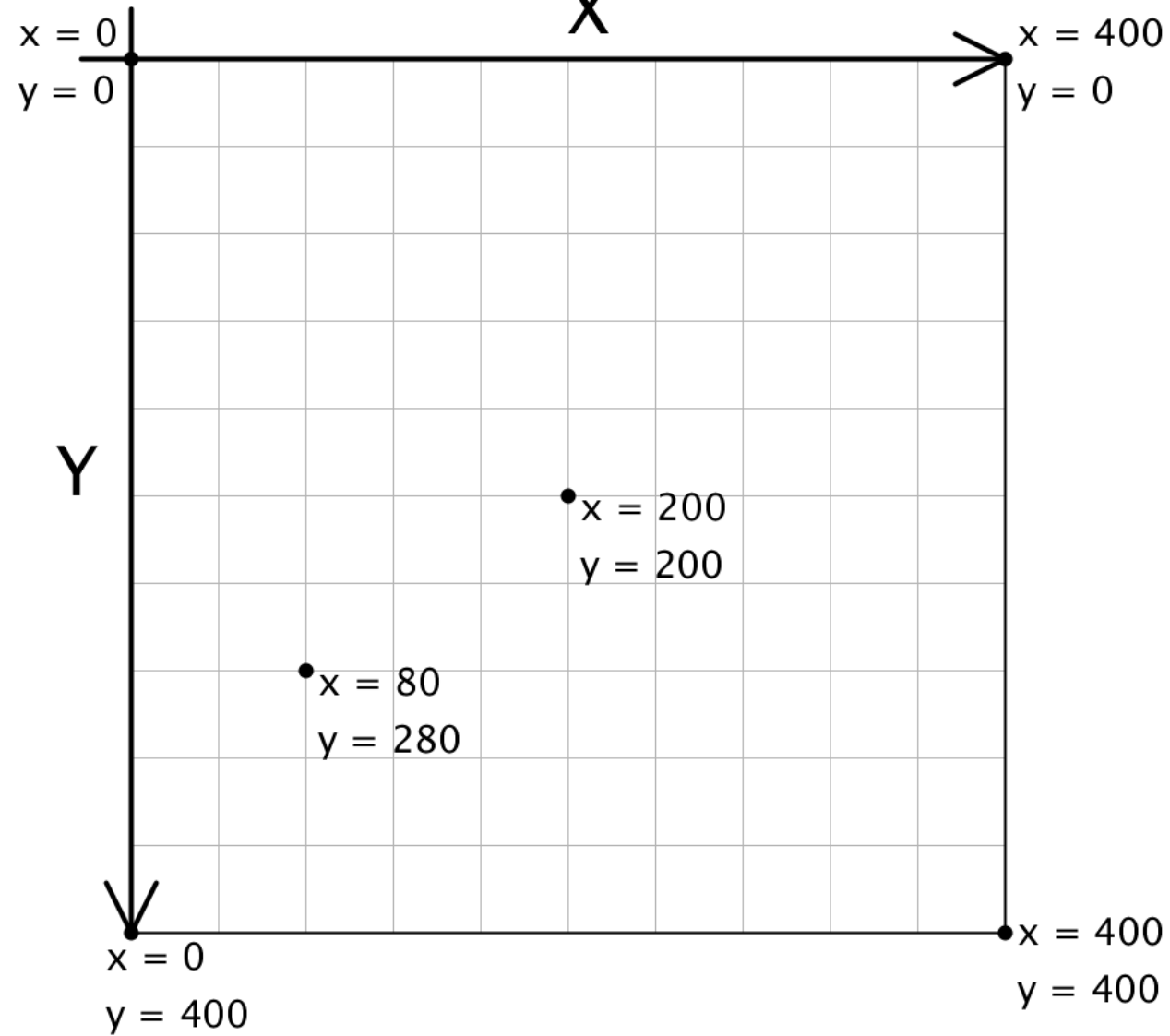
Structure of a Basic Sketch:

- **setup() function:** This is executed once when the sketch starts. It's commonly used to define initial environment properties such as screen size and background color.
 - **draw() function:** This runs repeatedly after setup(). It's used for continuously running code, such as animation or checking for input.
4. **Running the Sketch:** The sketch is compiled and run when you click the "Run" button (or press Ctrl+R/Cmd+R). The visual output is displayed in a separate window.

Enough Lecture: Today's IFOs

- You will get a sense of the PDE shortly
- How to **draw and color** with Processing
- What are **Variables** as simple as they get

Drawing in p5



Cartesian Coordinates in p5.js

Canvas Grid:

- Origin (0, 0) is at the top-left corner.
- X-axis increases to the right.
- Y-axis increases downward.

Using Coordinates:

- Position shapes using (x, y) coordinates.

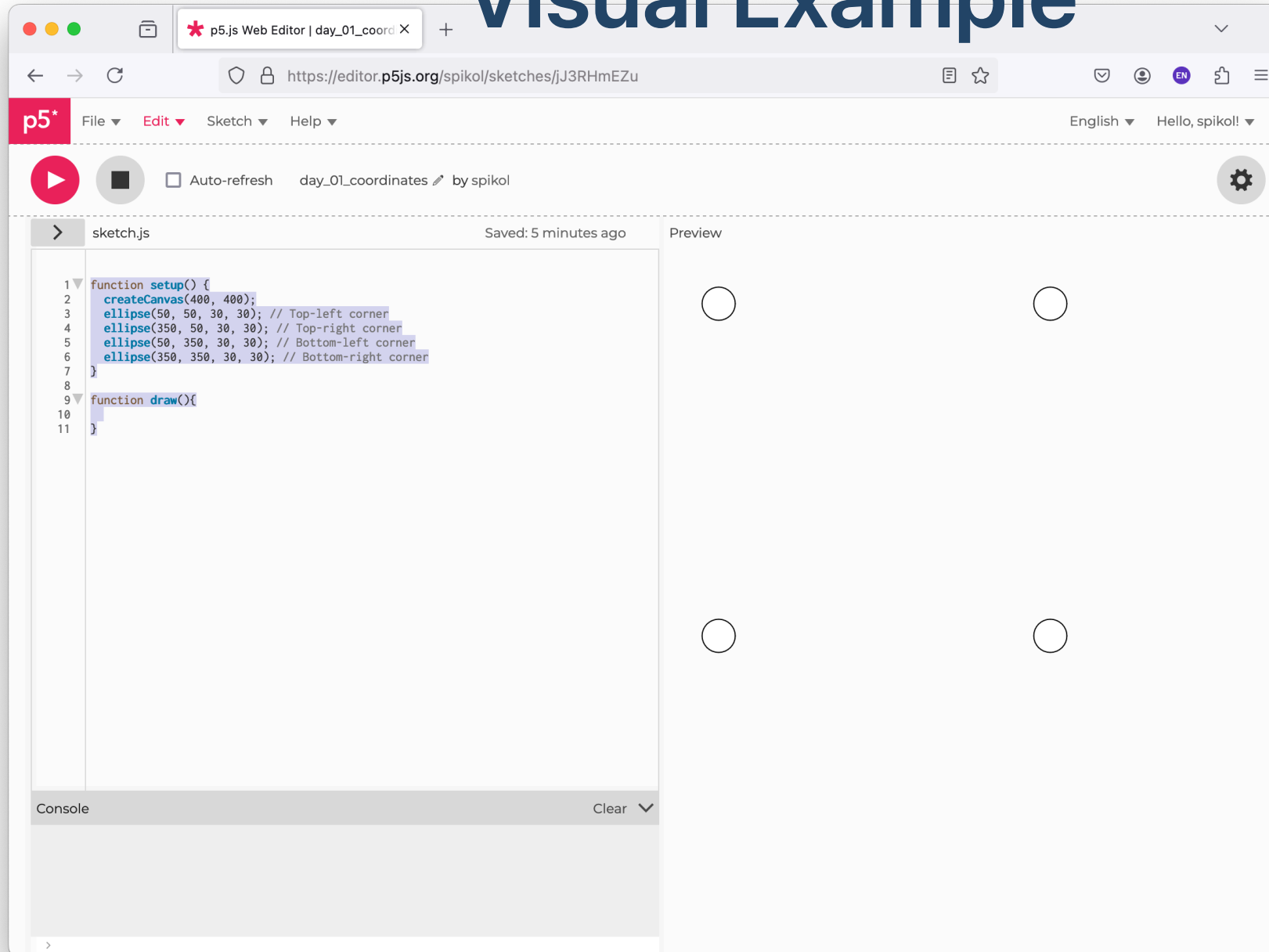
Example coordinates:

- (0, 0) - Top-left corner.
- (400, 400) - Bottom-right corner (for a 400x400 canvas).

Code Example

```
function setup() {  
  createCanvas(400, 400);  
  ellipse(50, 50, 30, 30); // Top-left corner  
  ellipse(350, 50, 30, 30); // Top-right corner  
  ellipse(50, 350, 30, 30); // Bottom-left corner  
  ellipse(350, 350, 30, 30); // Bottom-right corner  
}  
  
function draw(){  
  
}
```

Visual Example



Variables

- At its core, a **variable in programming** is like a container or a storage box that holds data.
- **Variables** are names that hold **values**.
- Imagine you have a box in real life where you put an apple; you might label that box "fruit" so you know it contains a fruit. In programming, the box is the variable, the apple is the data, and the label "fruit" is the variable's name.
- Keep in mind that, unlike real life, if you put an orange into your variable.

Scoping Variables

- **Global Variables:**
 - Declared outside any function.
 - Accessible from any part of the program.
- ****Local Scope**
 - Variables declared within function or blocks

